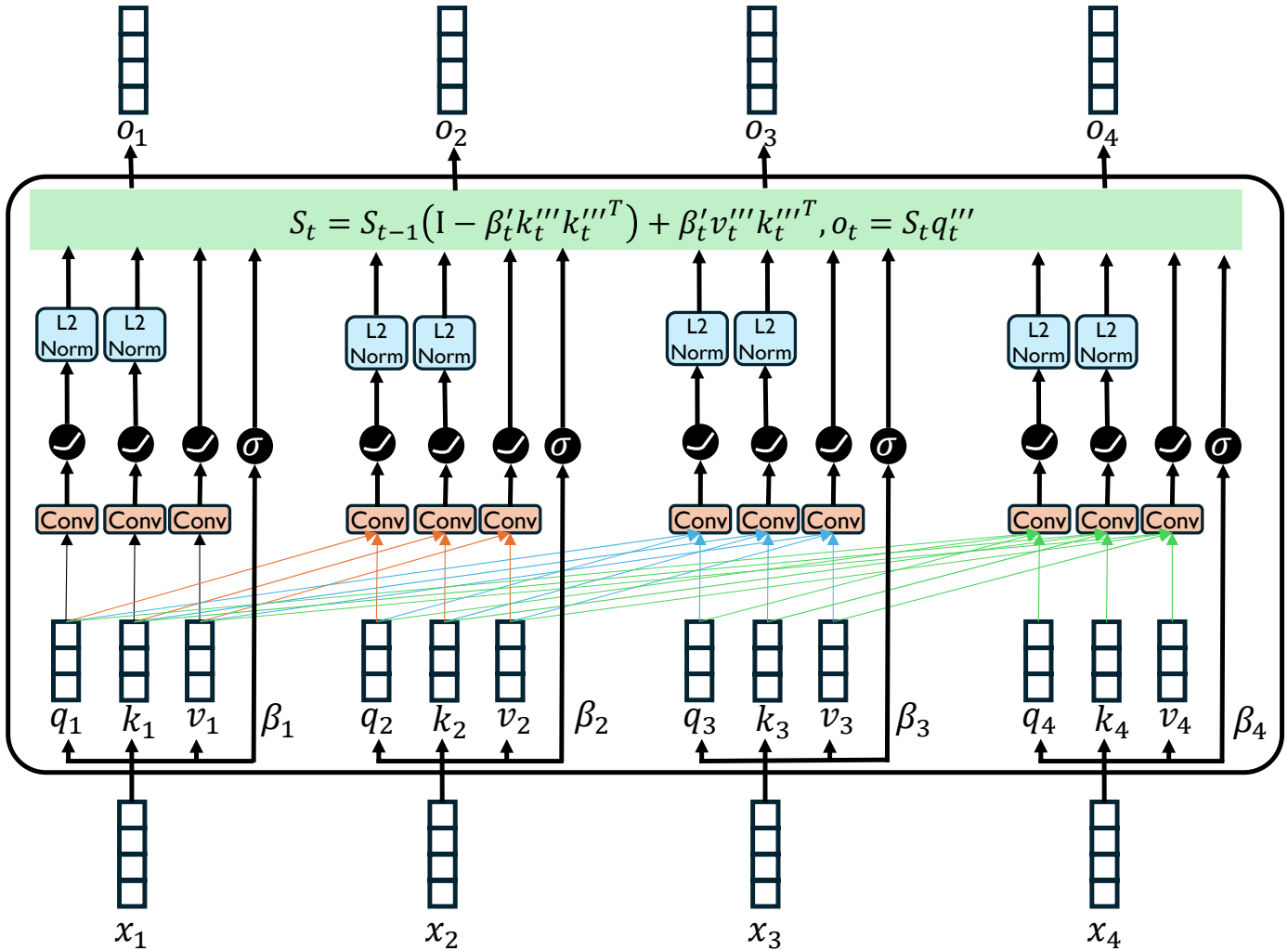


$$q'''_t = q''_t / ||q''_t||_2$$
$$k'''_t = k''_t / ||k''_t||_2$$
$$v'''_t = v''_t / ||v''_t||_2$$

$$q''_t = SiLU(q'_t) / ||\mathrm{SiLU}(q'_t)||_2$$
$$k''_t = SiLU(k'_t) / ||\mathrm{SiLU}(k'_t)||_2$$
$$v''_t = SiLU(v'_t) / ||\mathrm{SiLU}(v'_t)||_2$$
$$\beta'_t = Sigmoid(\beta_t)$$

$$q'_t = \sum_{j=t-3}^{t} w^Q_{Conv}[:, j] \odot q_j$$
$$k'_t = \sum_{j=t-3}^{t} w^K_{Conv}[:, j] \odot k_j$$
$$v'_t = \sum_{j=t-3}^{t} w^V_{Conv}[:, j] \odot v_j$$

$$q_t = W_Q x_t \ (W_Q \in R^{d_k \times d})$$
$$k_t = W_K x_t \ (W_K \in R^{d_k \times d})$$
$$v_t = W_V x_t \ (W_V \in R^{d_v \times d})$$
$$\beta_t = W_\beta x_t \ (W_\beta \in R^{1 \times d})$$